

SOFTWARE REQUIREMENTS SPECIFICATION

Teddy General Trading E-Commerce Website

Project Focus:	Showcasing, Managing, and Selling Water Materials & Related Equipment
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Prepared For:	Teddy General Trading (Client)
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1. Introduction

1.1 Purpose

The purpose of this document is to define the detailed, comprehensive software requirements for the Teddy General Trading E-Commerce Website. This document serves as a definitive, legally and technically binding guide for both the developer (Natnael Mamush) and the client (Teddy General Trading). It establishes mutual agreement regarding the system's expected features, functional capabilities, performance criteria, security parameters, and overall technical boundaries.

1.2 Project Scope

The scope of this project entails the complete end-to-end design, development, optimization, and deployment of a fully functional, high-performance e-commerce platform. The system is custom-engineered specifically for showcasing, marketing, managing inventory, and executing digital retail sales of **water materials and related industrial/commercial products** (such as water pumps, filtration systems, pipes, valves, and specialized fittings).

The deployment includes two distinct operational layers:

- **Customer-Facing Storefront:** A highly responsive, intuitive, and modern web catalog allowing users to discover products, filter by precise technical criteria, manage a digital shopping cart, and perform secure checkouts.
- **Administrative Backend Dashboard:** A secure corporate management portal allowing Teddy General Trading administrators full control over product catalogs, real-time inventory adjustments, order processing workflows, and site settings.

Critical Project Constraint

The core engineering lifecycle—from asset delivery to initial system deployment and testing—is strictly constrained to an accelerated execution window of **7 to 10 calendar days**. Advanced integrations such as live payment gateways are scheduled for post-deployment extension increments.

2. Overall Description

2.1 User Classes and Characteristics

The system distinguishes and caters to three primary roles, each configured with specific access rights and user journeys:

Guest / Visitor

An unauthenticated user visiting the website. Guests have access to browse the public product catalog, filter items by category, view individual item specifications, and submit general inquiries through the integrated "Contact Us" form. Guest users are explicitly required to register or sign in to complete a commercial purchase or checkout.

Registered Customer

An authenticated retail or wholesale customer. Registered customers enjoy full storefront privileges, including managing their personal customer profile, view historical orders, check real-time order status tracking, persist items within a virtual shopping cart, and execute the secure multi-step checkout workflow.

Administrator (Client)

Internal stakeholders representing Teddy General Trading. The administrator possesses complete, unrestricted access to the secure administrative backend dashboard. Privileges include comprehensive Product CRUD (Create, Read, Update, Delete) management, manual and automated stock level adjustments, order fulfillment tracking, customer data views, and static website content management.

2.2 Operating Environment

The software application is designed to behave reliably under the following technical conditions:

- **Hosting & Server Architecture:** Hosted on a locally implemented infrastructure environment utilizing an **Ethio Telecom Local Server Virtual Private Server (VPS)** setup, custom-configured with optimized server-side resource limits targeting high operational uptime.
- **Domain Routing:** Registered and managed through the **Namechamp** domain registrar, configured with proper DNS propagation, A-records, and naming structures pointing to the local host.
- **Client Platform Compatibility:** Developed as a fully responsive, modern web application optimized to be cross-browser compatible with all major modern web engines, including Google Chrome, Apple Safari, Mozilla Firefox, and Microsoft Edge. It guarantees visual and layout parity across Desktop, Tablet, and Mobile device Viewports.

3. System Features (Functional Requirements)

3.1 Product Catalogue Management

REQ-3.1.1: Administrative CRUD Logic

The system shall provide an intuitive administrative interface embedding explicit backend logic that allows the authenticated Administrator to Create, Read, Update, and Delete (CRUD) item listings dynamically via the dashboard.

REQ-3.1.2: Catalog Schema and Detailed Text Fields

Each distinct product listing shall comprehensively support the following baseline attributes: Title, Price (ETB), Category, Stock Quantity, and multiple product images. Crucially, the system shall provide a dedicated, unstructured free-form **Text Field** allowing the administrator to input comprehensive, custom technical details and specifications about the specific water material.

REQ-3.1.3: Storefront Display & Filtering Architecture

The customer-facing storefront shall display active products within a clean, performance-optimized, organized grid/list view. The module must natively support client-side/server-side pagination and instantaneous category filtering to ensure smooth navigation through extensive inventories.

3.2 Shopping Cart and Checkout System

REQ-3.2.1: Virtual Cart Mechanics

The storefront shall provide users with a persistent, dynamic virtual shopping cart. Users must have the ability to seamlessly add products, remove products, and adjust specific line-item quantities directly within the UI, instantly updating the local state.

REQ-3.2.2: Dynamic Cost Computation

The system shall automatically and dynamically recalculate item subtotals, any applicable taxes, shipping fees, and the absolute total cost in real-time as users modify cart states.

REQ-3.2.3: Checkout Ingestion

The checkout module shall securely collect customer shipping addresses, contact details, and billing declarations through validation-enforced web forms, ensuring data sanity before submitting payloads to the backend handlers.

3.3 Customer Account Management

REQ-3.3.1: Secure Account Provisioning

The application shall allow new users to register an account using a valid, unique email address and a custom secure password, enforcing standard complexity parameters during input validation.

REQ-3.3.2: Customer Activity Hub

Authenticated customers shall be granted immediate access to a personalized account dashboard where they can instantly track current order processing status, view complete historical purchases, and manage their address profiles.

REQ-3.3.3: Automated Credentials Recovery

The system shall provide a secure, automated "Forgot Password" recovery mechanism that safely transmits a time-sensitive, single-use cryptographic reset link or token to the customer's registered email address.

3.4 Payment Handling & Offline Processing Baseline

REQ-3.4.1: Baseline Order Status Log

For the core delivery phase, the system shall log orders with a state pending manual/offline verification (e.g., bank transfer or cash on delivery validations) prior to full digital payment API deployments.

3.5 Contact and Inquiry Forms

REQ-3.5.1: Structured Lead Capture

The system shall feature a public, dedicated "Contact Us" page containing a secure message form that accurately captures the inquirer's Full Name, Email Address, Subject, and specific textual Message payload.

REQ-3.5.2: Automatic SMTP E-mail Routing

Upon successful form validation and submission, the system's email subsystem shall immediately capture the payload, format it, and route it via SMTP to the Administrator's designated corporate email address for rapid response processing.

4. Non-Functional Requirements

4.1 UI/UX and Responsiveness

- **NFR-4.1.1: Fluid Viewport Adaptability:** The frontend user interface components must utilize fully responsive layout patterns, ensuring pixel-perfect layout preservation and seamless interactions across a broad spectrum of mobile smartphones, tablets, laptops, and wide-screen workstation monitors.
- **NFR-4.1.2: Brand Identity Cohesion:** The visual language, UI palette, typographic scales, button treatments, and iconography must strictly reflect the corporate brand identity of Teddy General Trading, utilizing designated color schemes and appropriate corporate logo placements.

4.2 Security & Authentication

- **NFR-4.2.1: Transport Layer Encryption:** The entire application domain must be forcefully served over Hypertext Transfer Protocol Secure (HTTPS) using a valid Secure Sockets Layer / Transport Layer Security (SSL/TLS) certificate to encrypt all data in transit.
- **NFR-4.2.2: Cryptographic Password Storage:** The database layer must never store plaintext credentials. Customer and administrative passwords must be safely salted and hashed using industry-standard, computationally slow cryptographic hashing algorithms (specifically **bcrypt**).
- **NFR-4.2.3: Role-Based Access Control (RBAC):** The administrative backend application route and API endpoints must be strictly protected via standard RBAC middleware, intercepting and terminating unauthenticated or non-admin requests immediately at the routing perimeter.

4.3 Search Engine Optimization (SEO)

- **NFR-4.3.1: Meta Tag Architecture:** The application architecture shall allow custom entry fields for basic on-page SEO optimization. This includes the administrative capability to inject unique, keyword-optimized Meta Titles, Meta Descriptions, and standard Image Alternative (Alt) attributes across all dynamic product pages.
- **NFR-4.3.2: Dynamic Sitemap Provisioning:** The backend routing framework shall maintain an automated engine that dynamically generates and structures a standardized XML sitemap index (`sitemap.xml`) at the domain root.

4.4 Performance & Reliability (SRE Metrics)

- **NFR-4.4.1: Speed Threshold:** Optimized first-contentful-paint and page load times shall not exceed 3 seconds when accessed under standard commercial broadband conditions.
- **NFR-4.4.2: Uptime Guarantee Target:** The selected target production environment (VPS or managed environment) must be contractually configured to support a 99.9% application uptime Service Level Agreement (SLA).

- **NFR-4.4.3: Automated Backup Continuity:** The persistence layer shall be supported by standard Cron automated scripting to ensure regular structural backups of database records.

5. Project Milestones & Core Deliverables

To hit the rapid **7 to 10 calendar day delivery timeline**, the core implementation framework focuses exclusively on baseline e-commerce operations, deferring external service integrations to post-launch phases:

Phase	Milestone Description	Estimated Timeline	Client Responsibility Matrix
1	Requirement Gathering & Asset Collection	Day 1	Deliver corporate logo vectors, full product inventory details, text data templates, and clear confirmation of 50% mobilization deposit.
2	Design & Local Environment Setup	Day 2 - 5	Provide comprehensive, prompt review cycles regarding initial design mockups, configure Namechamp domain pointers, and allocate Ethio Telecom VPS credentials.
3	Core E-Commerce & Admin CRUD Implementation	Day 6 - 8	Review the dynamic custom detail text fields and test structural administrator dashboard actions (Create, Update, Delete lists).
4	Testing, QA, and Security Audits	Day 9	Participate actively in structural User Acceptance Testing (UAT) for frontend catalogs and backend controls.
5	Final Deployment, Handover & Go-Live	Day 10	Execute final system sign-off, authorize local production deployment, and clear the outstanding 50% remaining contractual balance payment.

6. Post-Launch Roadmap (Increments & Product Extensions)

The following architectural expansions are officially recognized by both parties as outside the scope of the initial 7–10 day deployment cycle. These components are scheduled to be integrated as planned technical increments or feature extensions following successful go-live:

6.1 Phase II: Automated Payment Gateway Integration

- **API Webhook Engineering:** Future execution of complete automated API integrations with selected local/international payment platforms (such as **Telebirr, Chapa, or CBE Birr**) as chosen by the client [cite: 40].
- **Automated Order Automation:** Upgrading the system core to trigger automated server-side state transitions from "Pending Verification" to "Paid/Processing" instantly upon receipt of valid secure gateway payment confirmation webhooks [cite: 41].

6.2 Phase III: Artificial Intelligence (AI) Model Integration

- **AI-Powered Online Customer Support:** Deployment of an intelligent conversational AI agent/chatbot embedded within the storefront viewport to assist users with live inquiries, material support, and navigational aid.
- **Intelligent Catalog Organization & Search:** Implementation of semantic AI models designed to dynamically organize website modules, index item listings intuitively based on natural user search input, and provide robust automated identification of complex industrial water materials.

7. Maintenance and Support

Following the official launch and final deployment of the application on the production server, a standard **30-day post-launch warranty period** goes into immediate effect [cite: 68]. This warranty strictly covers critical bug fixes, server configuration patches, performance tuning, and minor interface text adjustments directly related to the baseline features outlined in Section 3 and 4 [cite: 68].

Any feature additions, major architectural revisions, core design overhauls, database schema expansions, or extended routine technical support beyond this 30-day window are explicitly excluded from this baseline project scope [cite: 69]. Such items, including the activation of components detailed in Section 6, will be handled under a distinct, mutually structured *Website Maintenance & Support Agreement* [cite: 69].